research. Projects such as the US Skyborg Program (2020), Boeing Airpower Teaming System (2020), and FCAS (Future Combat Air System, 2021) introduced the notion of cognitive autonomy, the ability of drones to perceive, process, and act in dynamic combat and environmental conditions with minimal human oversight. By the mid-2020s, these prototypes had matured into complex systems capable of collaborative decision-making, autonomous mission planning, and AI-based threat response, laying the groundwork for fifth-generation UAV evolution.

A significant catalyst in this transformation was the integration of quantum computing and edge AI, which addressed the limitations of onboard data processing in earlier generations. Quantum-enabled communication systems began allowing real-time, secure, and high-capacity data exchange across vast drone networks, facilitating coordinated missions involving hundreds of autonomous units. Simultaneously, digital twin technology, virtual replicas of UAVs capable of simulating missions and predicting system failures, emerged as a core feature of development, enabling continuous optimization and predictive maintenance.

The evolution of smart materials and nanotechnology further accelerated the transition toward fifth-generation drones. Innovations such as self-healing composites, morphing wings, and lightweight metamaterials allowed for enhanced stealth, flexibility, and resilience. Combined with hybrid propulsion systems including hydrogen fuel cells, solar-assisted energy modules, and silent electric turbines, these developments extended endurance while reducing logistical dependencies.

Parallel to technological advancements, the ethical and regulatory frameworks governing autonomous drones began to evolve. Global defense and aerospace organizations initiated discussions on AI ethics, autonomous decision accountability, and human–machine teaming (HMT) to ensure safe